

Social navigation with the Navel robot

Temporal social cues for rules and a language model



Background

A guide robot must recognise interest in interaction. Changing gaze and distance, and missed detections, make this decision difficult. [1]

Objective

Compare rules and a language model using the same temporal social state. Examine their action choices, cue availability and timely decision delivery.

Research question

How do rules and a language model respond to the same temporal social cues?

Methods

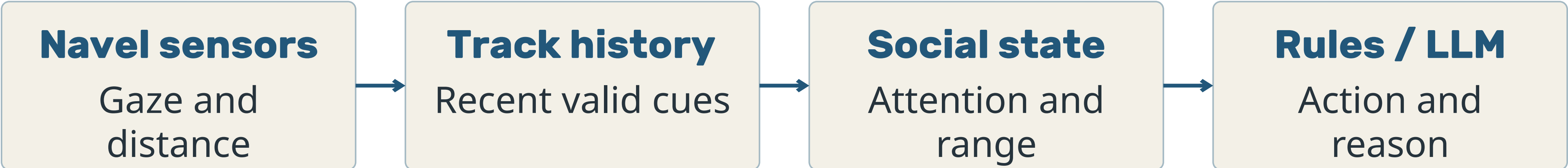


Figure 1. Nine SDK recordings feed both policies with identical initial eligible states. Rules and qwen2.5:7b return Continue, Approach, Engage or Yield, with a reason. [1]

Preliminary results

Initial survey

37 respondents chose the next action and rated each option (1–5). [2]

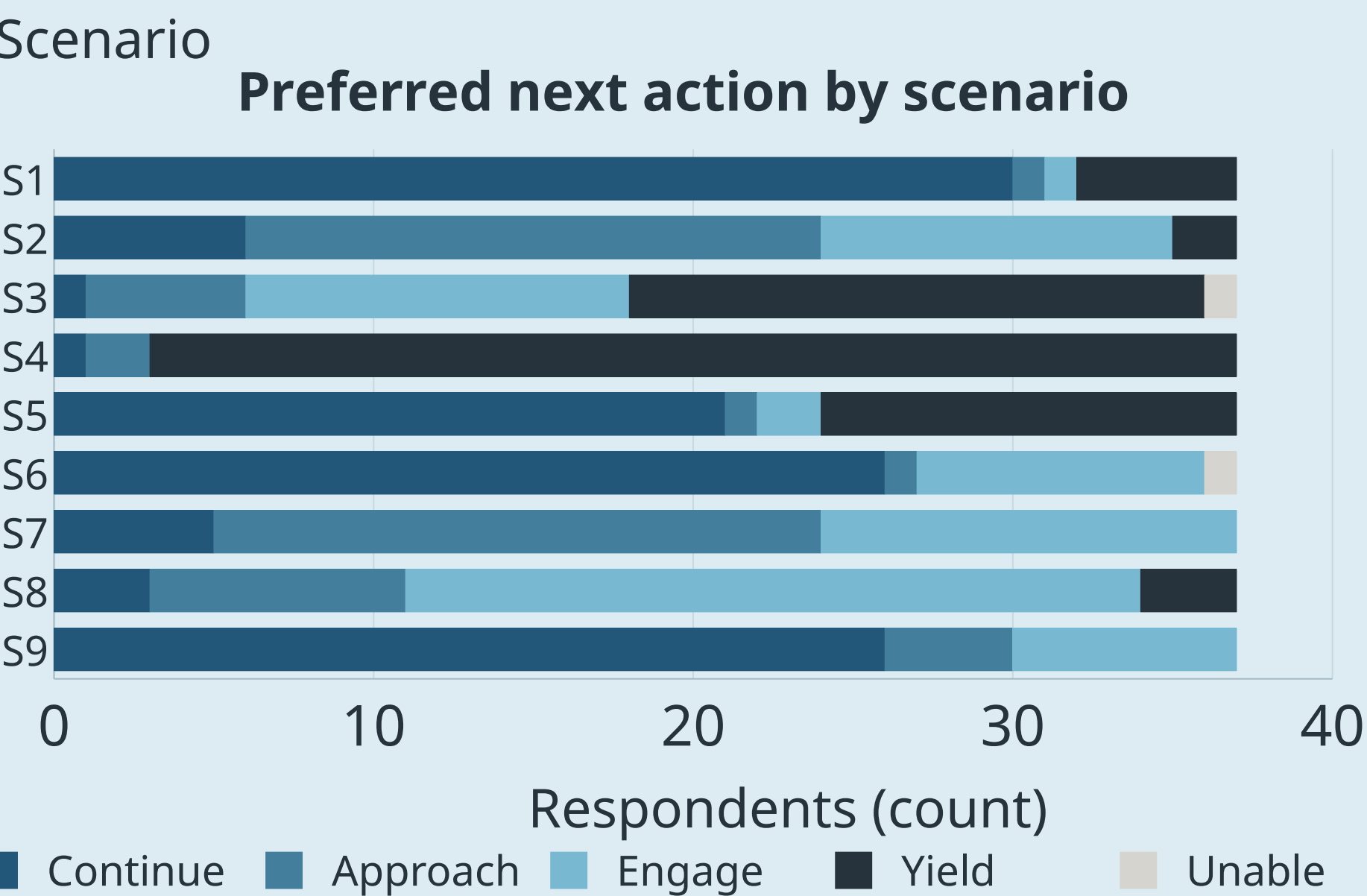


Figure 2. Every bar contains 37 responses. “Unable” means unable to judge.

Pipeline comparison

Outputs in 3/9 cases. Six lacked sufficient person or gaze evidence. [1]

Case	Rules	LLM
S2	Engage 8.28 s	Engage 12.39 s
S3	Engage 1.19 s	Engage 6.50 s
S9	Approach 1.22 s	Continue 8.90 s

Acceptance: rules 3/3, LLM 0/3.

Table 1. Time from replay start (s). No robot actions were executed.

Conclusion and next work

S4 favoured Yield (34/37). S3 split between Yield (18/37) and Engage (12/37). Replay exposes missing cues and delayed LLM delivery. Human-alignment accuracy awaits video alignment. Next: repeat trials and evaluate actual robot behaviour with people.